

[This question paper contains 5 printed pages.]

6628

Your Roll No.....

**B.Sc. (H) Computer Science / VI Sem. B**

**Paper 601 : Theory of Computation**

**(Admissions of 2001 and onwards)**

*Time : 3 Hours*

*Maximum Marks : 75*

*(Write your Roll No. on the top immediately  
on receipt of this question paper.)*

*Question No. 1 is compulsory. Attempt any  
four from the rest. Assume  $\Sigma = \{a, b\}$  for all the  
questions unless specified otherwise. Parts of a  
question must be answered together.*

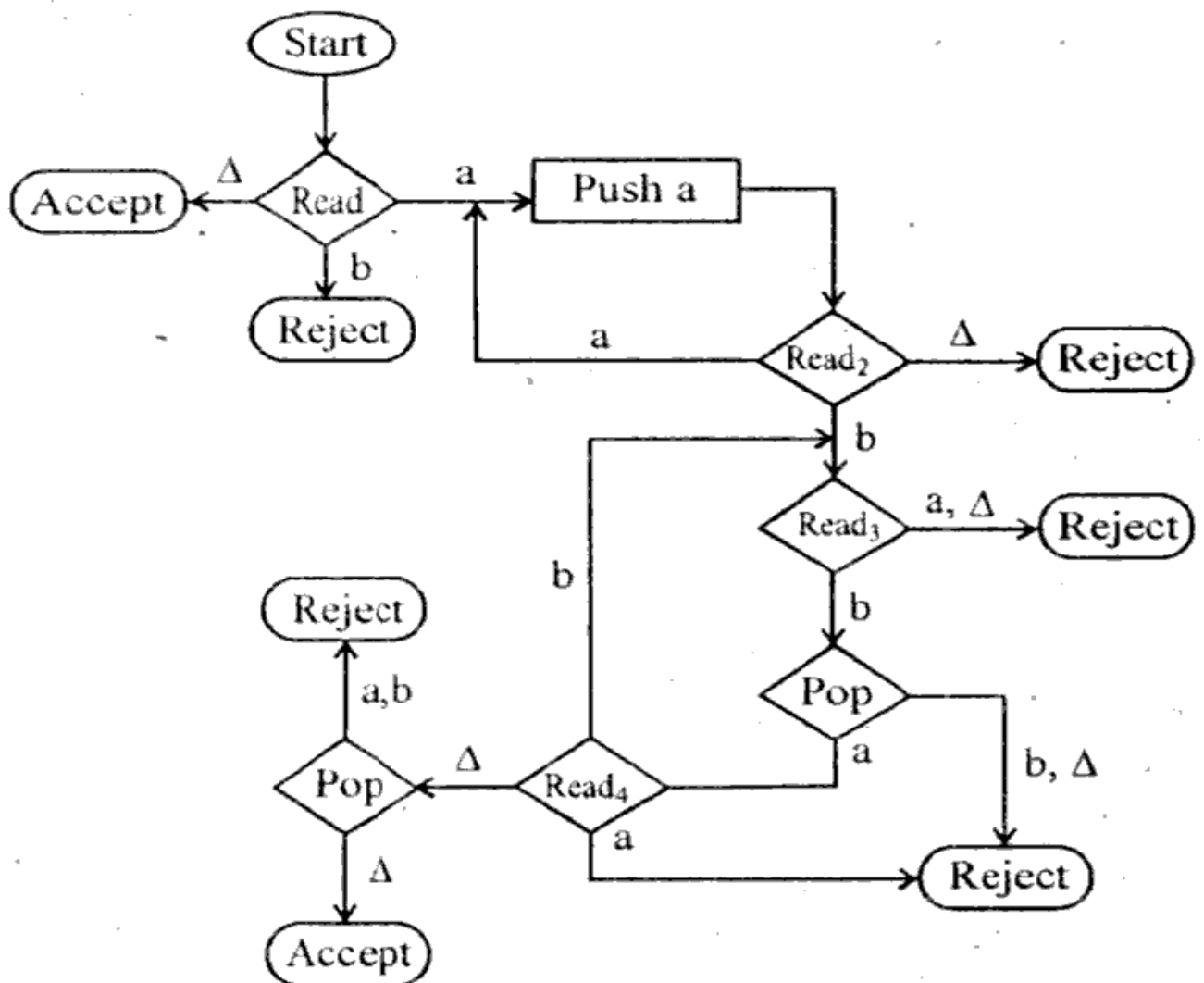
1. (a) What is a recursive language ? Illustrate with the help of an example. (4)
- (b) Write a regular expression for the set of all words that end in double letter. (3)
- (c) Give a grammar for the language of odd palindromes. (3)
- (d) Give FA that accepts all words with different first and last letters. (5)

P.T.O.

(e) Prove the language  $a^n b^n a^n$  for  $n = 1, 2, 3, \dots$  is non context-free. (5)

(f) Show that the CFG  $S \rightarrow aS \mid Sa \mid a$  is ambiguous. (4)

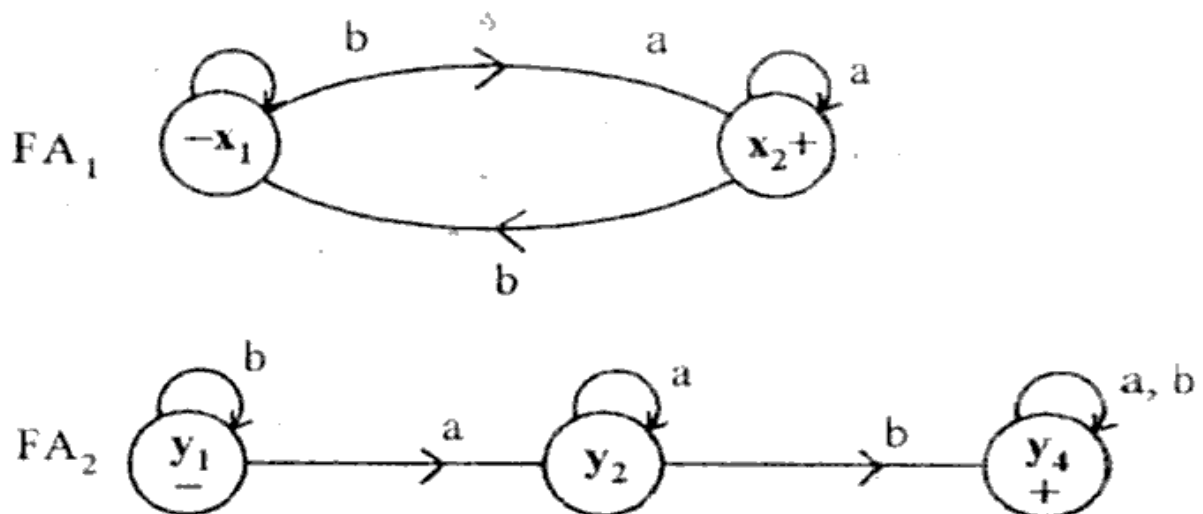
(g) What is the language accepted by the following PDA ? Find a CFG that generates this language. (4)



(h) State Kleene's Theorem. (2)

(i) Design a Turing machine that computes the function  $f(m) = m+2$ , where  $m$  is a positive binary number. (5)

2. (a) Show that  $(a + b)^* = (a + b)^*ab(a + b)^* + b^*a^*$ . (3)
- (b) Give regular expression for the language  $L$  where  $L = \{a^n b^m \mid m \geq 0, n \geq 3\}$ . (2)
- (c) If  $S = \{a, b\}$  and  $T^* = S^*$ . Prove that  $T$  must contain  $S$ . (3)
- (d) Describe the language associated with the regular expression  $(a(aa)^*b(bb)^*)^*$ . (2)
3. (a) Construct FA for language  $FA_1 FA_2$  where  $FA_1$  and  $FA_2$  are given below : (4)



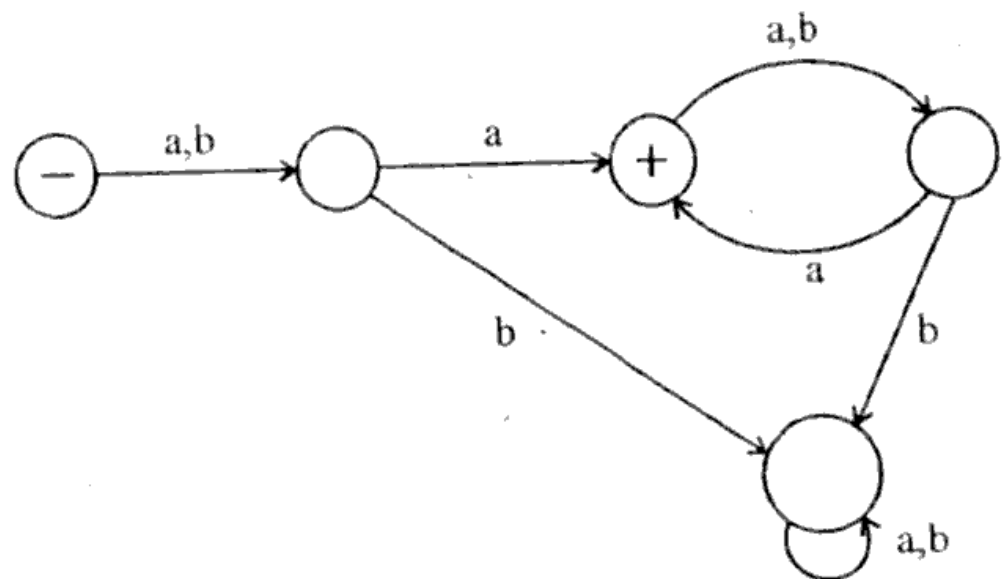
- (b) Construct Fas for the language  $L_1$  and  $L_2$  as given below.

$$L_1 = (a + b) b(a + b)^*, L_2 = b(a + b)^*$$

Find a regular expression and FA's that define  $L_1 \cap L_2$ . (6)

P.T.O.

4. (a) Using the bypass algorithm, convert the following TG into regular expression. (5)



- (b) Write a CFG for the language over the alphabet  $\{a, b\}$  consisting of all words with no three consecutive b's. (5)

5. (a) Draw a PDA for the language

$$\{a^n b^n a^m, \text{ where } n, m = 1, 2, 3, \dots\} \quad (5)$$

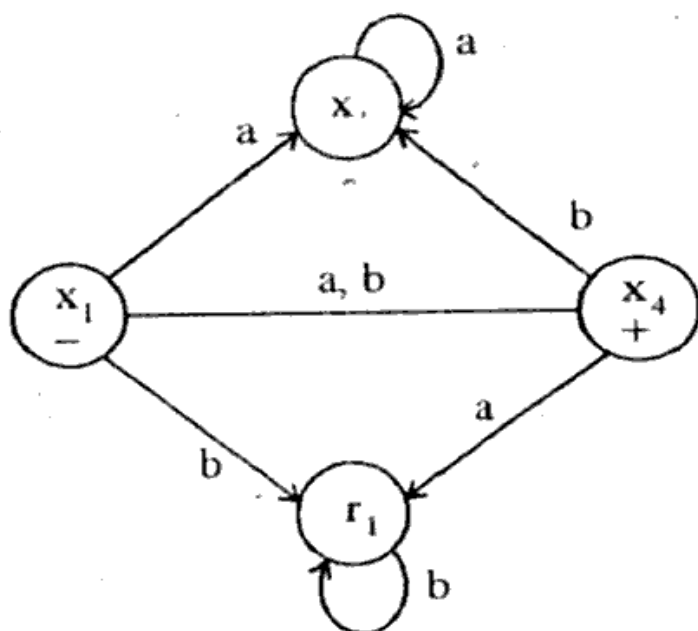
- (b) Show that the language

$$\{a^m b^n \text{ where } m = 2n + 1, n \geq 1\} \text{ is not regular.} \quad (5)$$

6. (a) Design a copying machine C which transform  $Uw\underline{U}$  into  $UwUw$  where w contain no blanks. (5)

- (b) Describe Universal Turing Machine with the help of suitable notations. (5)

7. (a) Convert the following NFA into DFA (5)



- (b) Are Context free languages closed under Intersection? Explain with example. (5)